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RULE · Basic probability (ehtimollik)

$$P(\text{event}) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$$

200 students, 80 prefer basketball: $P = \frac{80}{200} = \frac{2}{5} = 40\%$

From a table: total outcomes is the relevant row/column total – read the condition carefully before dividing.

RULE · Addition rule: $P(A$ or $B)$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

One die, even OR > 3: $\frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{2}{3}$

If A and B can't happen together, $P(A \text{ and } B) = 0$ and the rule reduces to $P(A) + P(B)$ – but always check for overlap first.

RULES SHEET – 2/2

RULE · Multiplication rule: independent events

$$P(A \text{ and } B) = P(A) \times P(B)$$

Two coin flips, heads then tails: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

Only valid when one event's outcome doesn't change the other's probability.

RULE · Mutually exclusive vs. independent

	Mutually exclusive	Independent
Question	Can they happen together at all?	Does one change the other's probability?
Condition	$P(A \text{ and } B) = 0$	$P(A \text{ and } B) = P(A)P(B)$
Rule used	Addition rule	Multiplication rule

If two events are mutually exclusive AND both have nonzero probability, they are automatically NOT independent.

One die: "rolling 3" and "rolling 4" are mutually exclusive but not independent – rolling a 3 makes $P(4)$ drop from $\frac{1}{6}$ to 0.